Relational Analysis (1)

CSE552 Program Analysis — Lecture 16

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Non-Relational Analysis

- Tracks each variable separately
- Ignores relationships between variables

```
x = input();
// x: [-inf, inf]
y = x;
// x: [-inf, inf], y: [-inf, inf]
z = x == y;
// x: [-inf, inf], y: [-inf, inf], z: [0, 1]
```

- Does not know that `x == y`, so `z` can be either 0 or 1

Relational Analysis

- Tracks relationships between variables

```
x = input();
y = x;
// x = y
z = x == y;
// x = y, z = 1
```

- Knows that `x == y`, so `z` is always 1

Motivation 1: Dynamically Sized Arrays

```
x = input();
y = [0; x];
if z < x {
  y[z] = 1;
}
```

- We need to know that `z < x` to ensure that `z` is a valid index

Motivation 2: Binary Analysis

```
x = sp - 4;
*x = 0;

y = sp - 4;
z = *y;
```

- Local variables are accessed relatively to the stack pointer (`sp`)
- We need to identify addresses relative to `sp` to track the values of local variables

Abstract Domain of Linear Equalities[1]

- $Var = \{x_i \mid i \in I\}$
- $State = \{\bot\} \cup \left\{ \bigwedge_{i \in I} \left(b_i = \sum_{j \in I} a_{i,j} x_j \right) \right\}$
- If the program has n variables, use n linear equalities
- An equality can be ignored if its every coefficient is 0
  - Effectively, we have at most n equalities

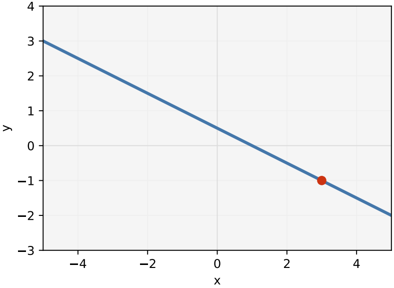
[1] Affine relationships among variables of a program (Karr, 1976)

2-Dimensional Linear Equalities

- $Var = \{x, y\}$
- $State =$
  $\{\bot\} \cup \{\top\}$
  $\cup \{b_1 = a_{1,1}x + a_{1,2}y\}$
  $\cup \{b_1 = a_{1,1}x + a_{1,2}y$
  $\wedge\, b_2 = a_{2,1}x + a_{2,2}y\}$

- No equalities: plane
- One equality: line
- Two equalities: point

- $x + 2y = 1$
- $x + 2y = 1 \wedge x - y = 4$



3-Dimensional Linear Equalities — Definition

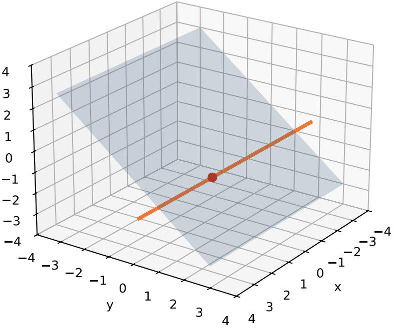
- $Var = \{x, y, z\}$
- $State =$

$$
\begin{aligned}
&\{\bot\} \cup \{\top\} \cup \{b_1 = a_{1,1}x + a_{1,2}y + a_{1,3}z\} \\
&\cup \{b_1 = a_{1,1}x + a_{1,2}y + a_{1,3}z \\
&\qquad \wedge b_2 = a_{2,1}x + a_{2,2}y + a_{2,3}z\} \\
&\cup \{b_1 = a_{1,1}x + a_{1,2}y + a_{1,3}z \\
&\qquad \wedge b_2 = a_{2,1}x + a_{2,2}y + a_{2,3}z \\
&\qquad \wedge b_3 = a_{3,1}x + a_{3,2}y + a_{3,3}z\}
\end{aligned}
$$

- No equalities: space
- One equality: plane
- Two equalities: line
- Three equalities: point

3-Dimensional Linear Equalities — Example

- $y + z = 0$
- $y + z = 0 \wedge y - z = 2$
- $y + z = 0 \wedge y - z = 2 \wedge x = 1$



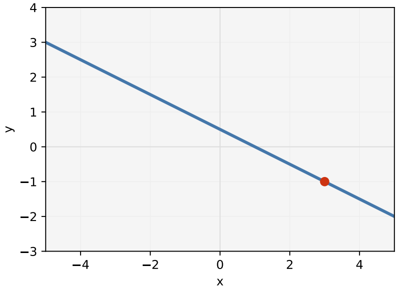
State Representation: Matrix

- Matrix representation: $AX = B$
  - where X is a vector made of the program variables

Matrix Representation: 2-Dimensional Examples

- $x + 2y = 1$
$$\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1$$

- $x + 2y = 1 \wedge x - y = 4$
$$\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$



Matrix Representation: 3-Dimensional Examples

- $y + z = 0$

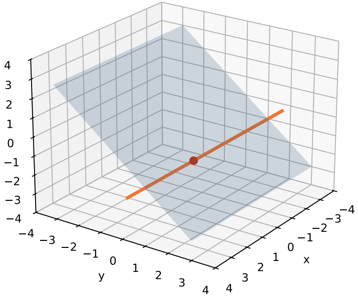
$$\begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

- $y + z = 0 \wedge y - z = 2$

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

- $y + z = 0 \wedge y - z = 2 \wedge x = 1$

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$



State Representation: Geometric

- A point P and a generating vector basis $U_0, \ldots, U_m$
  - where $P + \sum_{i=0}^{m} \mu_i U_i$ constitute the set of solutions

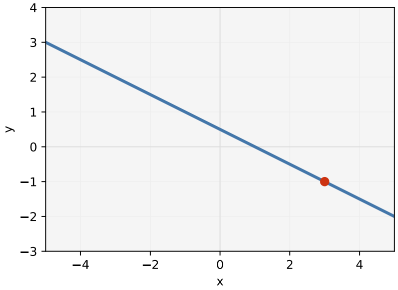
Geometric Representation: 2-Dimensional Examples

- $x + 2y = 1$

  $P = (1, 0), \quad U_0 = (-2, 1)$

- $x + 2y = 1 \wedge x - y = 4$

  $P = (3, -1)$



Geometric Representation: 3-Dimensional Examples

- $y + z = 0$

  $P = (0, 0, 0)$

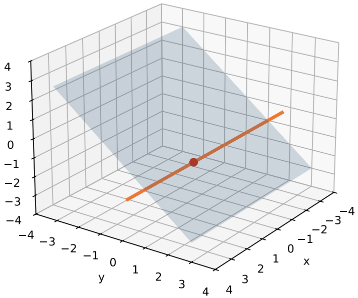
  $U_0 = (1, 0, 0), \quad U_1 = (0, 1, -1)$

- $y + z = 0 \wedge y - z = 2$

  $P = (0, 1, -1), \quad U_0 = (1, 0, 0)$

- $y + z = 0 \wedge y - z = 2 \wedge x = 1$

  $P = (1, 1, -1)$

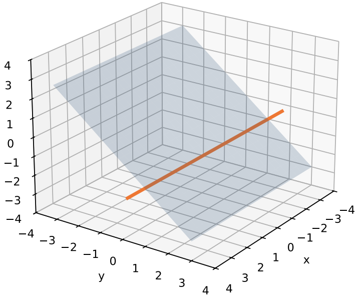


Order

- $s_1 \subseteq s_2$ if and only if the solution set of s_1 is a subset of the solution set of s_2
- Where s_1 is represented by $P, U_0, \ldots, U_m$ and s_2 is represented by A' and B', check if $A'P = B'$ and $A'U_i = 0$ for all i

Order — Example 1

- $s_1 : y + z = 0 \wedge y - z = 2$
  - $P = (0, 1, -1)$
  - $U_0 = (1, 0, 0)$
- $s_2 : y + z = 0$
  - $A' = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}$
  - $B' = 0$



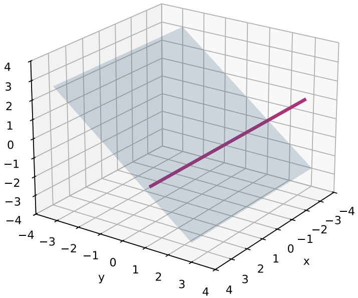
$$A'P = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = 0 = B'$$

$$A'U_0 = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

Order — Example 2

- $s_1 : y + z = 2 \wedge y - z = 2$
  - $P = (0, 2, 0)$
  - $U_0 = (1, 0, 0)$
- $s_2 : y + z = 0$
  - $A' = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}$
  - $B' = 0$

$$A'P = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = 2 \neq B'$$

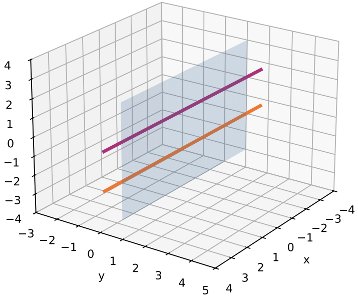


Join

- Where s_1 is represented by $P, U_0, \ldots, U_m$ and s_2 is represented by $P', U'_0, \ldots, U'_n$, use P as the point and normalize $P' - P, U_0, \ldots, U_m, U'_0, \ldots, U'_n$ for the generating vector basis

Join — Example

- $s_1 : y + z = 0 \wedge y - z = 2$
  - $P = (0, 1, -1)$
  - $U_0 = (1, 0, 0)$
- $s_2 : y + z = 2 \wedge y - z = 0$
  - $P' = (0, 1, 1)$
  - $U_0' = (1, 0, 0)$
- $P'' = (0, 1, -1)$
- $P' - P = (0, 0, 2)$, $U_0 = (1, 0, 0)$, $U_0' = (1, 0, 0)$
  - $\Rightarrow U_0'' = (1, 0, 0), U_1'' = (0, 0, 1)$
- $s_1 \sqcup s_2 : y = 1$



Widening

- The lattice has a finite height of $|Var| + 1$

$$\begin{array}{c}
\text{No equalities} \\
| \\
1 \text{ equality} \\
| \\
\vdots \\
| \\
|Var| - 1 \text{ equalities} \\
| \\
|Var| \text{ equalities} \\
| \\
\bot
\end{array}$$

- Just use the join operator as a widening operator

Control-Sensitivity

- If a condition is defined by a linear equality, we add the equality to the state and normalize it
- If a condition is defined by a linear inequality, we cannot add a new equality but may produce $\perp$ if it is unsatisfiable

Control-Sensitivity — Example

Current state: $x + 2y = 1$

- Condition: $2x + 4y = 2$
  - $\Rightarrow x + 2y = 1$
- Condition: $x - y = 4$
  - $\Rightarrow x = 3 \wedge y = -1$
- Condition: $x + 2y = 2$
  - $\Rightarrow \perp$

Assignments (Invertible)

- $x_o = b + \sum_{i \in I} a_i x_i$ where $a_o \neq 0$
  - $x_o^{\text{new}} = b + a_o x_o^{\text{old}} + \sum_{i \in I \setminus \{o\}} a_i x_i$
  - $x_o^{\text{old}} = \frac{1}{a_o}\left(x_o^{\text{new}} - b - \sum_{i \in I \setminus \{o\}} a_i x_i\right)$
- Update the state by substituting x_o with $\frac{1}{a_o}\left(x_o - b - \sum_{i \in I \setminus \{o\}} a_i x_i\right)$

Invertible Assignment — Example

- Current state: $x + y = 1$
- Assignment: `x = x + 1;`
- Updated state: $x + y = 2$

Assignments (Non-Invertible)

- $x_o = b + \sum_{i \in I} a_i x_i$ where $a_o = 0$
- Normalize the basis so that nonzero coefficient for x_o occurs in at most one equality
- Drop the equality with nonzero coefficient for x_o
- Add the new equality $x_o = b + \sum_{i \in I} a_i x_i$

Non-Invertible Assignment — Example

- Current state: $x + y = 0 \wedge x + z = 0$
- Assignment: `x = y - z;`
- Normalized: $x + y = 0 \wedge y - z = 0$
- Updated state: $x - y + z = 0 \wedge y - z = 0$
  or $x = 0 \wedge y - z = 0$

Putting It Together

```
x = 1
// x = 1
if ... {
  y = 2
  // x = 1, y = 2
} else {
  y = 4
  // x = 1, y = 4
  x = x + 1
  // x = 2, y = 4
}
// 2x - y = 0
if x + 2 == y {
  // x = 2, y = 4
} else {
  // 2x - y = 0
}
```

Abstract Domain of Linear Inequalities[2]

- $Var = \{x_i \mid i \in I\}$
- $State = \{\bot\} \cup \left\{\bigwedge_i \left(\sum_{j \in I} a_{i,j} x_j \leq b_i\right)\right\}$
- Use an arbitrary number of linear inequalities
- Subsume interval domain
  - $x : [l, h] \iff x \leq h \wedge -x \leq -l$
- Subsume linear equalities
  - $e = b \iff e \leq b \wedge -e \leq -b$

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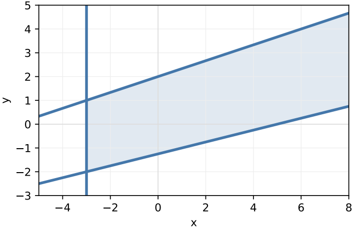
[2] Automatic discovery of linear restraints among variables of a program (Cousot and Halbwachs, 1978)

2-Dimensional Linear Inequalities

- $Var = \{x, y\}$
- $-3 \leq x \wedge 3y \leq x + 6 \wedge x - 5 \leq 4y$



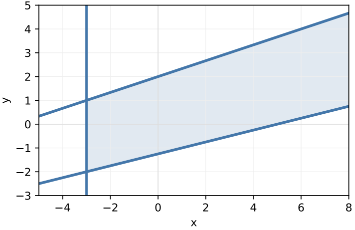
State Representation: Matrix

- Matrix representation: $AX \leq B$
  - where X is a vector made of the program variables
  - $\leq$ is component-wise

Matrix Representation: Example

- $Var = \{x, y\}$
- $-3 \leq x \wedge 3y \leq x + 6 \wedge x - 5 \leq 4y$

$$\begin{pmatrix} -1 & 0 \\ -1 & 3 \\ 1 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \leq \begin{pmatrix} 3 \\ 6 \\ 5 \end{pmatrix}$$



State Representation: Geometric

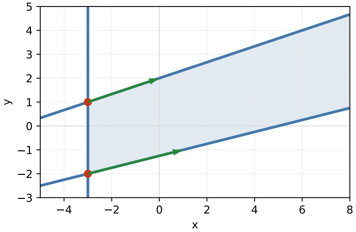
- A set of points $P_0, \ldots, P_n$ and a set of vectors $U_0, \ldots, U_m$
  - where $\left\{ \sum_{i=0}^{n} \lambda_i P_i + \sum_{j=0}^{m} \mu_j U_j \;\middle|\; \sum_{i=0}^{n} \lambda_i = 1 \wedge \lambda_i, \mu_j \geq 0 \right\}$ is the set of solutions

Geometric Representation: Example

- $Var = \{x, y\}$
- $-3 \leq x \wedge 3y \leq x + 6 \wedge x - 5 \leq 4y$

$$P_0 = \begin{pmatrix} -3 \\ -2 \end{pmatrix}, \quad P_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$U_0 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad U_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$



Order

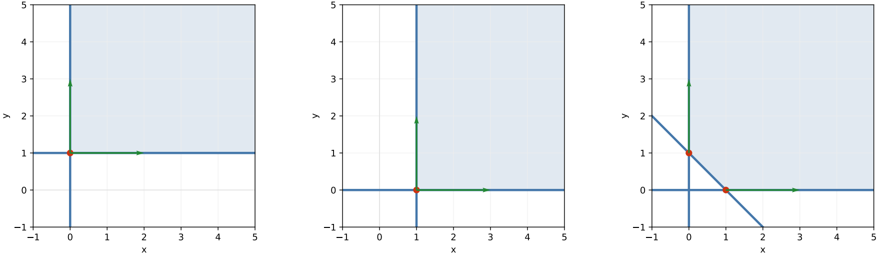
- $s_1 \subseteq s_2$ if and only if the solution set of s_1 is a subset of the solution set of s_2
- Where s_1 is represented by $P_0, \ldots, P_n, U_0, \ldots, U_m$ and s_2 is represented by A' and B', check if $A'P_i \leq B'$ for all i and $A'U_j \leq 0$ for all j

Join

- Where s_1 is represented by $P_0, \ldots, P_n, U_0, \ldots, U_m$ and s_2 is represented by $P'_0, \ldots, P'_n, U'_0, \ldots, U'_m$, use $P_0, \ldots, P_n, P'_0, \ldots, P'_n$ as the points and $U_0, \ldots, U_m, U'_0, \ldots, U'_m$ as the vectors, but normalize them

Join — Example

- s_1: $P_0 = (0,1), U_0 = (0,1), U_1 = (1,0)$
- s_2: $P'_0 = (1,0), U'_0 = (0,1), U'_1 = (1,0)$
- $s_1 \sqcup s_2$: $P''_0 = (1,0), P''_1 = (0,1), U''_0 = (1,0), U''_1 = (0,1)$

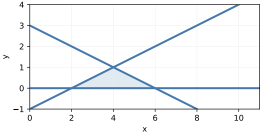


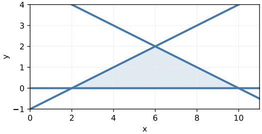
Widening

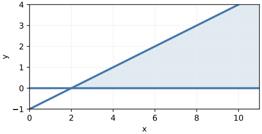
- The lattice has an infinite height
- Same principle as for intervals
  - Keep stable inequalities

Widening — Example

- s_1: $2y \leq x - 2 \land 2y \leq -x + 6 \land y \geq 0$
- s_2: $2y \leq x - 2 \land 2y \leq -x + 10 \land y \geq 0$
- $s_1 \sqcup s_2$: $2y \leq x - 2 \land y \geq 0$







Control-Sensitivity

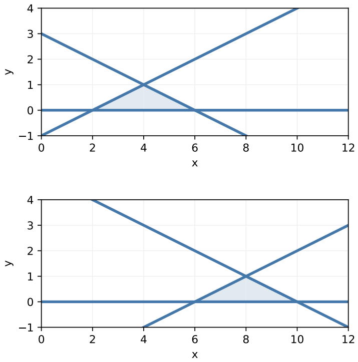
- If a condition is defined by a linear (in)equality, we add it to the state and normalize it
- Produce $\perp$ if unsatisfiable

Assignments

- Invertible: $x_o = b + \sum_{i \in I} a_i x_i$ where $a_o \neq 0$
  - Update the state by substituting x_o with $\frac{1}{a_o}\left(x_o - b - \sum_{i \in I \setminus \{o\}} a_i x_i\right)$
- Non-invertible: $x_o = b + \sum_{i \in I} a_i x_i$ where $a_o = 0$
  - Drop inequalities on x_o and add new inequalities

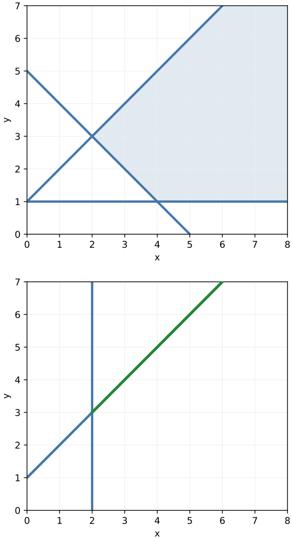
Invertible Assignment — Example

- Current state:
  $2y \leq x - 2 \wedge 2y \leq -x + 6 \wedge y \geq 0$
- Assignment: `x = x + 4;`
- Updated state:
  $2y \leq x - 6 \wedge 2y \leq -x + 10 \wedge y \geq 0$



Non-Invertible Assignment — Example

- Current state:
  $y \geq 1 \wedge y \geq -x + 5 \wedge y \leq x + 1$
- Assignment: `y = x + 1;`
- Normalized: $x \geq 2$
- Updated state:
  $x \geq 2 \wedge y \geq x + 1 \wedge y \leq x + 1$



Summary

- Relational analyses track relationships among variables
- The abstract domain of linear equalities represents states as conjunctions of linear equalities
- The abstract domain of linear inequalities represents states as conjunctions of linear inequalities, subsumes intervals and equalities